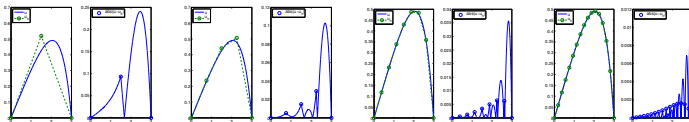


MA378 Chapter 4: Finite Element Methods

§4.3 Analysis: Cea's Lemma

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Start: 27 March 2026 (W11.2)



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3.1 Error analysis

Recall that we wrote the differential equation

$$-u''(x) + r(x)u(x) = f(x) \quad \text{on } (a, b), \quad u(a) = u(b) = 0,$$

in a variational form:

Define $A(u, v) := (u', v') + (ru, v)$. Find $u \in H_0^1(a, b)$ such that

$$A(u, v) = (f, v) \quad \text{for all } v \in H_0^1(a, b). \quad (1)$$

3.1 Error analysis

We then defined the **FEM**:

Definition 3.1 (The Finite Element Method)

Let S be a finite dimensional subspace of $H_0^1(a, b)$. The **Galerkin Finite Element method** is: *find $u_h \in S$ such that*

$$\mathcal{A}(u_h, v_h) = (f, v_h) \quad \text{for all} \quad v_h(x) \in S. \quad (2)$$

We now show that the member of S found by the FEM is the “closest” to the true solution.

3.2 Cea's Lemma

Lemma 3.2 (Cea's Lemma; Thm 14.6 of Süli and Mayers)

Let u be the solution to (1), i.e., the true solution, and let u_h be the solution to (2), i.e., the FE approximation.

- (i) The difference between the true and approximate solutions is orthogonal to S , i.e.,

$$\mathcal{A}(u - u_h, v_h) = 0 \text{ for all } v_h \in S,$$

and

- (ii) There is no element of S that is closer to u than u_h :

$$\mathcal{A}(u - u_h, u - u_h) = \min_{v_h \in S} \mathcal{A}(u - v_h, u - v_h),$$

3.2 Cea's Lemma

First we will prove that

$$A(u - u_h, v_h) = 0 \text{ for all } v_h \in S,$$

(which is a property known as **Galerkin Orthogonality**).

We have that $A(u, v) = (f, v)$ for all $v \in H_0^1(a, b)$
& that $u \in H_0^1(a, b)$. Also we have that
 $u_h \in S$ & $A(u_h, v_h) = (f, v_h) \quad \forall v_h \in S$.

But, since $S \subseteq H_0^1(a, b)$, we also have
 $A(u, v_h) = (f, v_h) \quad \forall v_h \in S$.

Subtracting we get

$$A(u, v_h) - A(u_h, v_h) = \underbrace{(f, v_h) - (f, v_h)}_{=0}.$$

3.2 Cea's Lemma

First we will prove that

$$A(u - u_h, v_h) = 0 \text{ for all } v_h \in S,$$

(which is a property known as **Galerkin Orthogonality**).

So this gives

$$A(u - u_h, v_h) = 0$$

as required.

Next we will prove that

$$\mathcal{A}(u - u_h, u - u_h) \leq \mathcal{A}(u - v_h, u - v_h) \text{ for any } v_h \in S.$$

$$\begin{aligned} \mathcal{A}(u - v_h, u - v_h) &= \mathcal{A}(\underbrace{u - u_h}_{\textcircled{1}} + \underbrace{u_h - v_h}_{\textcircled{2}}, \underbrace{u - u_h}_{\textcircled{3}} + \underbrace{u_h - v_h}_{\textcircled{4}}) \\ &= \mathcal{A}(\underbrace{u - u_h}_{\textcircled{1}}, \underbrace{u - u_h}_{\textcircled{3}}) + \mathcal{A}(\underbrace{u - u_h}_{\textcircled{1}}, \underbrace{u_h - v_h}_{\textcircled{4}}) + \\ &\quad \mathcal{A}(\underbrace{u_h - v_h}_{\textcircled{2}}, \underbrace{u - u_h}_{\textcircled{3}}) + \mathcal{A}(\underbrace{u_h - v_h}_{\textcircled{2}}, \underbrace{u_h - v_h}_{\textcircled{4}}) \\ &\geq \mathcal{A}(u - u_h, u - u_h) \quad \text{because } \mathcal{A}(u_h - v_h, u_h - v_h) \geq 0, \\ &\quad \text{and } \mathcal{A}(u - u_h, u_h - v_h) = 0 \text{ since } u_h - v_h \in S. \end{aligned}$$

3.2 Cea's Lemma

Since $\mathcal{A}(\cdot, \cdot)$ is an inner product (see Definition 3.6.1) it induces a **norm**:

$$|||u||| := \sqrt{\mathcal{A}(u, u)}.$$

So we can write (ii) of Cea's Lemma as

$$|||u - u_h||| \leq |||u - v_h||| \quad \text{for all } v_h \in S.$$

3.3 Numerical Analysis

With a little more work we can show that

$$|||u - u_h||| \leq C h ||u''||_{\infty}.$$

$$e \quad |||u - u_h||| \leq C h^{-1} ||u''||_{\infty}$$

That is, the error (in the energy norm) is proportional to h . We can then further deduce that the method converges:

$$\lim_{h \rightarrow 0} |||u - u_h||| = 0.$$

We'll sketch the analysis. The core is

- ▶ Let l be the piecewise linear spline interpolant to u , exactly as as described in Section 2.1.
- ▶ Note that l belongs to S .
- ▶ So, u_h as at least as good an approximation to u as l . That is

$$|||u - u_h||| \leq |||u - l|||.$$

- ▶ So now we need to estimate $|||u - l|||$. We know how to do that!

3.3 Numerical Analysis

First, note that if (for simplicity) $r \equiv 1$, then

$$|||u|||^2 = \|u'\|_2^2 + \|u\|_2^2.$$

Cauchy's theorem told us that on $[x_i, x_{i+1}]$,

$$u(x) - l(x) = \frac{u''(\tau)}{2}(x - x_i)(x - x_{i+1}).$$

Integrate $(u(x) - l(x))^2$ on $[x_i, x_{i+1}]$ to get

$$\int_{x_i}^{x_{i+1}} (u(x) - l(x))^2 dx = \frac{u''(\tau)h^5}{120}. \Rightarrow \|u - l\|_{2, [x_i, x_{i+1}]}$$

Then sum over all N intervals, and take the square root to get $\sim h^{5/2}$

$$\|u - l\|_2 \leq \frac{\|u''\|_\infty}{120} h^2,$$

3.3 Numerical Analysis

At first, it might seem strange that our error estimate depends on u'' . And, since we don't know u , we certainly don't know u'' either.

However, we can get a very sharp bound for it. In Exercise 1.4 of Section 4.1 you would have seen that we can apply a Maximum Principle to get an upper bound for u .

Now we apply the differential equation to get that

$$\|u''\|_{\infty} = \|-f + r(x)u(x)\|_{\infty},$$

which we can say is bounded by some constant.

We can now conclude that there is a constant C_0 such that

$$\|u - l\|_2 \leq C_0 h^2.$$

In the same way, we can also show that there is a constant, C_1 such that

$$\|u' - l'\|_2 \leq C_1 h.$$

3.3 Numerical Analysis

Putting all this together we get

$$|||u - u_h||| \leq |||u - l||| \leq C_0 h^2 + C_1 h \leq Ch.$$

And that's it!

We will now **demonstrate numerically** that the method converges... *and that this is sharp.*

The table opposite shows the maximum error, over all mesh points, in the finite element solution to

$$-u'' + 3u = x \text{ on } (0, 3),$$

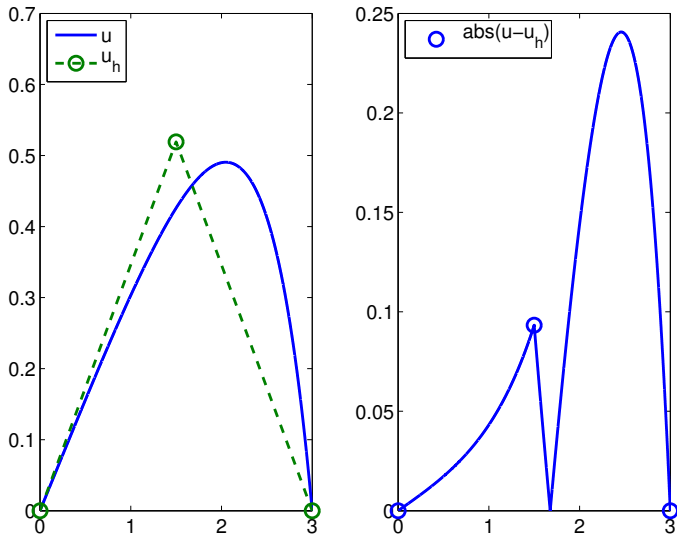
$$u(0) = u(3) = 0.$$

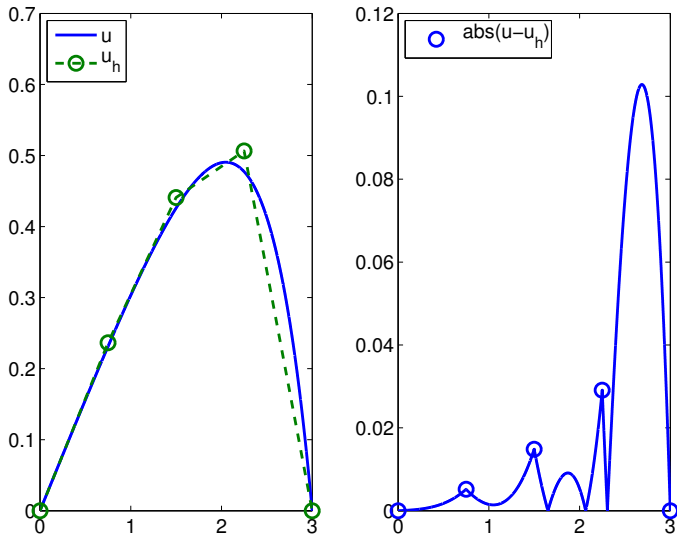
One can see that the error is proportional to N^{-1} (and thus to h).

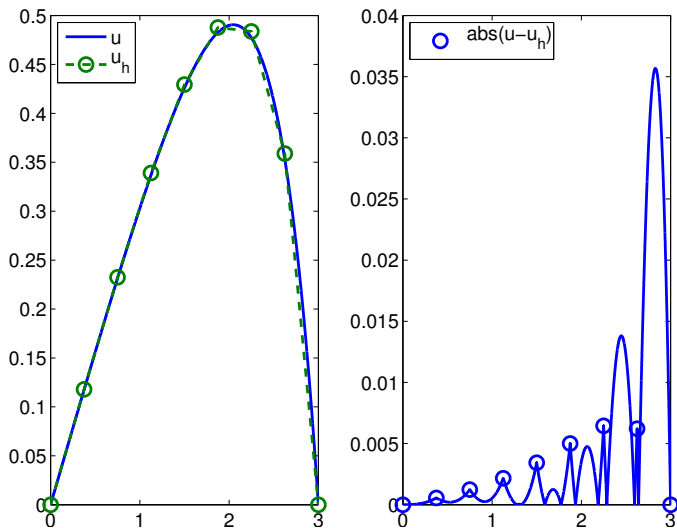
N	$ u - u_h $
8	1.721e-01
16	8.691e-02
32	4.357e-02
64	2.180e-02
128	1.090e-02
256	5.451e-03
512	2.726e-03
1024	1.363e-03

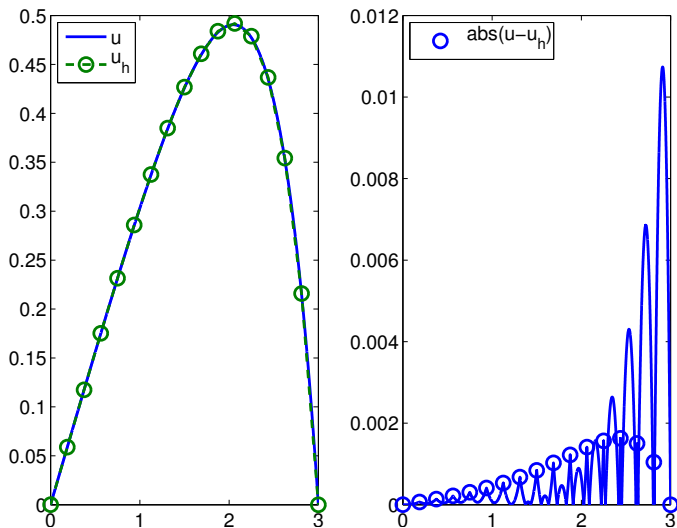
These results were generated in a Python/Firedrake. You can access the notebook on Colab at

https://colab.research.google.com/drive/1fQH9WMwRjsGEBicOW4fMRHvQ_efoLi6V?usp=sharing

Solution and error with $n = 2$ 

Solution and error with $n = 4$ 

Solution and error with $n = 8$ 

Solution and error with $n = 16$ 

3.4 FE Wrap-Up

There are many aspects of finite element methods that we did not cover, including

1. There are many **other** choices of basis functions given here. One could use cubic splines, or, indeed, higher-order polynomials.
2. When we try to improve the accuracy of the method by reducing h where the error seems large. This is called a h -FEM (and is the most common type of adaptive method).
3. We can also try to improve the accuracy of the method by increasing the order of the polynomials. This is called a p -FEM.
4. The ideas presented here extend to far more general problems. In particular, they work very well for problems in higher dimensions, and on weird-shaped domains.

3.5 Exercises

Exercise 3.1

Suppose that we want to solve

$$-u''(x) + u'(x) = 1 \text{ on } (a, b),$$

- (a) Write down the system of linear equations that we would have to solve in terms of h .
- (b) Explain why the analysis of Lemma 3.2 does not apply directly to this problem.

Exercise 3.2

Show that, for any function $f \in C^2[a, b]$,

$$\|f\|_2 \leq \sqrt{b-a} \|f\|_\infty,$$

where $\|f\|_2 := \left(\int_a^b (f(x))^2 dx \right)^{1/2} = \sqrt{(f, f)}$, and $\|f\|_\infty := \max_{a \leq x \leq b} |f(x)|$.